

WASP-35b

R-1429

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_211112 · created 2026-08-01T07:15:23+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459530.869 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.106 +/- 0.012
Transit depth (Rp/Rs)^2	1.11 +/- 0.26 [%]
Orbital Inclination (inc)	87.09 +/- 0.96
Airmass coefficient 1 (a1)	1.2361 +/- 0.0085
Airmass coefficient 2 (a2)	-0.144 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.68 %
Best Comparison Star	None
Optimal Aperture	5.5
Optimal Annulus	22.69
Transit Duration (day)	0.1227 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.106 ± 0.012 vs 0.1241 (deviation 0.78σ)
Duration fit vs published	0.1227 d vs 0.1317 d (deviation 0.63σ)
Mid-time O–C	7.1 min
Depth SNR	4.5
Residual scatter	0.68 %

Light curve

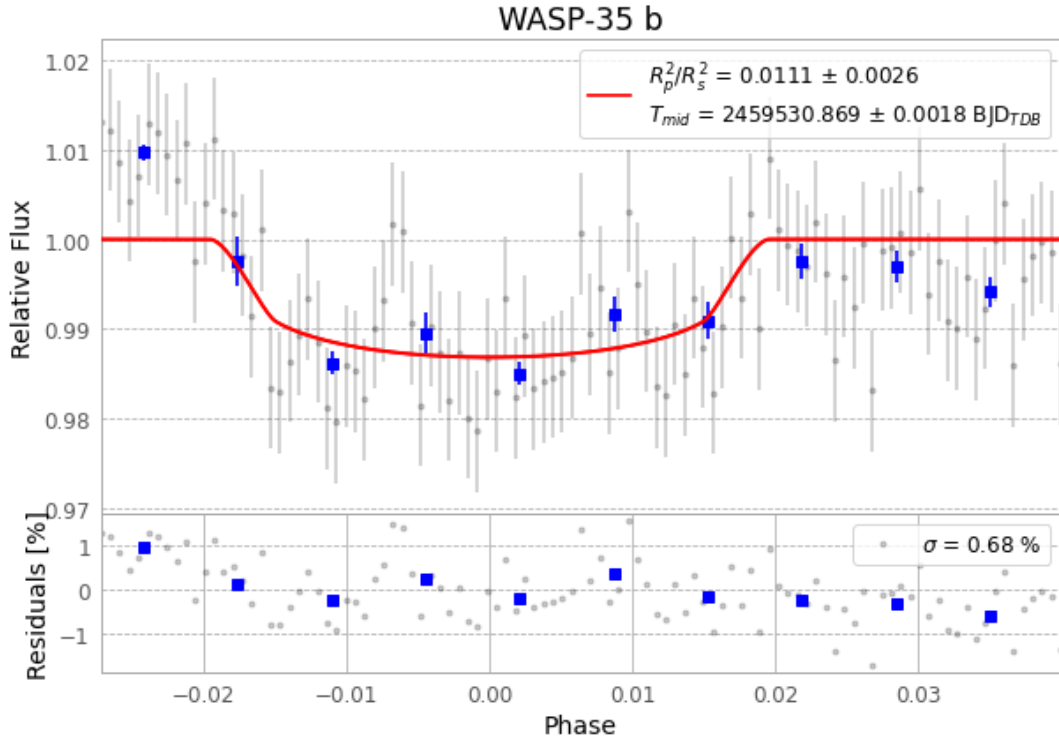


Figure 1 — FinalLightCurve_WASP-35 b_2021-11-11.png (SHA-256 d43e5d3bf3cdf05...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [445, 148], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 641b1d95f0061e64...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), WASP-35211112063912.FITS → WASP-35211112114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-211112.log (9b97bb267980...), FinalLightCurve_WASP-35 b_2021-11-11.png (d43e5d3bf3cd...), FinalLightCurve_WASP-35 b_2021-11-11.csv (e971147030f4...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 07:15Z.